

MOKOPANE, South AfricaWMO: **682710**Lat: **24.1967S**Lon: **29.0067E**Elev: **3599**StdP: **12.88**Time Zone: **2.00 (E02)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.0	43.2	16.7	14.6	59.2	20.4	17.5	59.8	16.5	60.8	12.1	63.5	1.7	70	0.296

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.1	94.0	64.4	91.6	64.4	89.4	64.4	70.7	81.8	69.8	80.8	69.0	79.9	5.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	118.4	73.6	67.1	114.2	73.2	66.1	110.0	72.9	37.0	82.0	36.2	80.7	35.5	79.9	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	11.1	9.5	DB	36.7	98.8	3.6	2.1	34.2	100.3	32.1	101.5	30.1	102.7	27.5	104.3	(4)
(5)			WB	31.1	71.9	2.7	1.3	29.2	72.8	27.6	73.6	26.1	74.3	24.1	75.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.7	75.8	76.4	74.6	69.5	64.1	59.3	58.5	64.9	71.3	73.6	74.1	74.7	(6)	
	DBStd	7.74	3.99	3.78	3.93	4.61	4.29	3.69	3.69	5.24	5.87	5.93	5.51	4.64	(7)	
	HDD50	1	0	0	0	0	0	0	0	1	0	0	0	0	(8)	
	HDD65	551	0	0	0	12	67	175	203	67	15	6	5	1	(9)	
	CDD50	7186	801	739	763	585	436	280	263	461	638	730	723	767	(10)	
	CDD65	2262	336	319	298	147	38	4	1	63	203	272	278	303	(11)	
	CDH74	17022	2446	2222	1863	832	350	79	62	601	1712	2352	2252	2251	(12)	
	CDH80	5985	878	808	603	215	51	4	2	126	647	957	857	837	(13)	
(14) Wind	WSAvg	4.2	4.2	4.2	4.2	3.8	3.4	3.3	3.1	4.5	5.1	5.4	5.1	4.3	(14)	
(15) Precipitation	PrecAvg	19.6	3.5	3.2	2.0	1.0	0.4	0.2	0.1	0.1	0.2	1.5	3.9	3.6	(15)	
	PrecMax	30.8	7.8	8.5	4.9	3.7	1.5	1.3	0.6	0.4	1.5	3.1	10.7	5.4	(16)	
	PrecMin	12.1	1.6	0.2	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	1.5	(17)	
	PrecStd	5.1	1.7	2.3	1.3	0.9	0.5	0.3	0.2	0.1	0.3	0.8	2.2	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.7	94.3	93.1	89.4	84.7	79.8	79.0	86.4	93.6	96.7	96.4	95.8	(19)	
		MCWB	65.7	66.5	65.1	64.1	58.8	55.2	53.4	58.0	59.9	63.1	64.0	67.0	(20)	
	2%	DB	90.8	90.8	89.2	85.4	80.9	76.1	75.3	83.5	90.1	92.5	92.2	91.5	(21)	
		MCWB	66.8	66.4	65.7	62.7	57.8	54.0	53.1	56.9	59.5	62.6	63.9	66.3	(22)	
	5%	DB	88.0	87.9	86.2	82.1	78.0	73.1	72.6	80.4	87.1	89.1	88.6	88.2	(23)	
		MCWB	66.9	66.6	65.5	62.1	57.4	53.4	52.7	55.4	58.9	62.0	64.1	67.0	(24)	
	10%	DB	85.1	85.2	83.3	79.0	74.9	70.4	69.4	77.1	83.6	85.6	85.1	84.7	(25)	
		MCWB	67.0	66.5	65.4	61.6	56.8	52.8	51.9	54.9	58.1	61.6	64.0	66.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.1	71.7	70.7	68.0	62.9	58.3	57.2	61.1	63.9	67.5	69.9	71.1	(27)	
		MCDB	82.2	82.0	81.7	78.9	76.0	72.2	69.4	77.7	84.9	87.0	82.6	85.2	(28)	
	2%	WB	70.7	70.4	69.4	66.3	61.1	56.6	55.7	59.4	62.5	65.9	68.5	69.9	(29)	
		MCDB	80.8	80.8	80.1	77.8	73.4	70.0	69.0	76.6	81.3	83.0	81.3	82.7	(30)	
	5%	WB	69.6	69.5	68.4	65.0	59.7	55.4	54.3	58.0	61.4	64.7	67.3	68.9	(31)	
		MCDB	79.6	79.9	78.7	76.6	72.4	68.4	68.0	74.7	79.8	81.0	80.0	81.1	(32)	
	10%	WB	68.7	68.5	67.4	63.7	58.4	54.1	53.0	56.6	60.3	63.7	66.2	68.0	(33)	
		MCDB	78.7	78.9	77.5	74.9	71.3	67.0	66.5	72.4	78.0	79.5	78.5	79.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	18.1	18.4	19.9	25.0	26.1	25.2	25.0	23.7	21.5	19.6	17.8	(35)	
		MCDBR	23.2	22.7	22.9	24.5	28.5	29.7	28.7	28.6	27.5	26.2	24.6	23.8	(36)	
	5% WB	MCWBR	6.9	6.1	6.5	7.3	10.4	12.4	12.3	10.9	9.5	8.3	7.1	6.7	(37)	
		MCWBR	17.1	18.0	17.4	20.2	23.1	24.9	23.7	23.9	23.3	21.7	19.8	18.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.385	0.376	0.348	0.310	0.294	0.290	0.347	0.401	0.425	0.390	0.390	(40)	
		taud	2.376	2.377	2.388	2.429	2.490	2.513	2.509	2.334	2.214	2.189	2.334	2.359	(41)	
	Ebn at Noon		304	299	293	288	287	286	292	283	281	284	300	303	(42)	
		Edh at Noon	42	41	38	34	29	27	28	36	45	49	43	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		2098	1999	1793	1543	1442	1282	1382	1623	1885	2021	2073	2068	(44)	
	RadStd		166	183	158	130	60	77	35	75	101	135	179	139	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page